

DBMS – WEEK 6

Product Review and Rating Management System

Project Name: ArtHub – An Online Art Marketplace

1. Objective

To develop a Product Review and Rating Management System for ArtHub using SQL. The system is designed to store customer feedback, manage artwork ratings, retrieve reviews, calculate average ratings, and identify highly rated artworks and artworks requiring improvement.

2. Scenario

ArtHub is an online art marketplace that offers various artworks, including paintings, portraits, digital art, sculptures, and handmade crafts. Customer feedback plays an important role in understanding customer satisfaction and evaluating artwork quality.

The Product Review and Rating Management System is connected to the existing Customer and ArtProduct tables. Customers can provide ratings from 1 to 5 stars and write comments about purchased artworks. The system stores these reviews and helps analyze artwork performance through rating and feedback reports.

3. Create Review and Rating Table

The Review table is created to store customer feedback and ratings for different artworks. Each review contains a unique review ID, customer ID, artwork ID, review date, rating, and review comments.

The table maintains a relationship with the existing Customer and ArtProduct tables. This relationship ensures that reviews are associated with valid customers and artworks.

```
SQL> CREATE TABLE Review (Review_ID VARCHAR2(10), Customer_ID NUMBER, Artwork_ID NUMBER, Review_Date DATE, Rating NUMBER, Review_Comments VARCHAR2(500));  
Table created.
```

4. Define Relationships and Constraints

Constraints are used to maintain the accuracy and integrity of the review data.

- **Primary Key:** Review_ID uniquely identifies each review.

```
SQL> ALTER TABLE Review ADD CONSTRAINT pk_review PRIMARY KEY (Review_ID);  
Table altered.
```

- **Foreign Key:** Customer_ID connects the Review table with the Customer table.

```
SQL> ALTER TABLE Review ADD CONSTRAINT fk_review_customer FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID);  
Table altered.
```

- **Foreign Key:** Artwork_ID connects the Review table with the ArtProduct table.

```
SQL> ALTER TABLE Review ADD CONSTRAINT fk_review_artwork FOREIGN KEY (Artwork_ID) REFERENCES ArtProduct(Artwork_ID);  
Table altered.
```

- **CHECK Constraint:** Ensures that customer ratings remain between 1 and 5.
- **NOT NULL Constraint:** Ensures that essential review details are provided.

```
SQL> ALTER TABLE Review ADD CONSTRAINT chk_review_rating CHECK (Rating BETWEEN 1 AND 5);
Table altered.

SQL> ALTER TABLE Review MODIFY Customer_ID NUMBER NOT NULL;
Table altered.

SQL> ALTER TABLE Review MODIFY Artwork_ID VARCHAR2(10) NOT NULL;
Table altered.

SQL> ALTER TABLE Review MODIFY Rating NUMBER NOT NULL;
Table altered.

SQL> ALTER TABLE Review MODIFY Review_Date DATE NOT NULL;
Table altered.

SQL> ALTER TABLE Review MODIFY Review_Comments VARCHAR2(500) NOT NULL;
Table altered.
```

These constraints help prevent invalid records and maintain consistency across the database tables.

5. Store Customer Feedback and Ratings

Customer feedback and ratings are stored in the Review table. Each review is associated with a customer and a specific artwork. Customers can express their opinions through written comments and assign a rating based on their satisfaction.

Sample reviews are added for different artworks available in ArtHub. The ratings range from 1 to 5 stars, representing different levels of customer satisfaction.

The system also supports updating review comments when corrections are necessary and deleting inappropriate or invalid reviews when required.

```
SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R1', 1, '101', TO_DATE('25-09-2026','DD-MM-YYYY'), 5, 'Beautiful artwork with excellent colors');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R2', 2, '102', TO_DATE('26-09-2026','DD-MM-YYYY'), 4, 'Good artwork with attractive colors');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R3', 3, '103', TO_DATE('27-09-2026','DD-MM-YYYY'), 3, 'Good portrait with decent detailing');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R4', 4, '104', TO_DATE('28-09-2026','DD-MM-YYYY'), 5, 'Excellent charcoal artwork with fine details');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R5', 5, '105', TO_DATE('29-09-2026','DD-MM-YYYY'), 4, 'Creative digital artwork with vibrant colors');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R6', 6, '106', TO_DATE('30-09-2026','DD-MM-YYYY'), 2, 'Artwork is good but needs more detailing');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R7', 7, '107', TO_DATE('01-10-2026','DD-MM-YYYY'), 5, 'Beautiful painting with a peaceful atmosphere');
1 row created.

SQL> INSERT INTO Review (Review_ID, Customer_ID, Artwork_ID, Review_Date, Rating, Review_Comments) VALUES ('R8', 8, '108', TO_DATE('02-10-2026','DD-MM-YYYY'), 4, 'Impressive sculpture with detailed craftsmanship');
1 row created.
```

6. Retrieve Product Review Details

The system provides several ways to retrieve customer feedback and review information.

Display All Reviews for a Product: Retrieves all reviews associated with a particular artwork, including its rating and comments.

```
SQL> SELECT * FROM Review;
```

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

R1	1	101	25-SEP-26	5
----	---	-----	-----------	---

R2	2	102	26-SEP-26	4
----	---	-----	-----------	---

R3	3	103	27-SEP-26	3
----	---	-----	-----------	---

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

R4	4	104	28-SEP-26	5
----	---	-----	-----------	---

R5	5	105	29-SEP-26	4
----	---	-----	-----------	---

R6	6	106	30-SEP-26	2
----	---	-----	-----------	---

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
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REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

R7	7	107	01-OCT-26	5
----	---	-----	-----------	---

R8	8	108	02-OCT-26	4
----	---	-----	-----------	---

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

R4	4	104	28-SEP-26	5
----	---	-----	-----------	---

R5	5	105	29-SEP-26	4
----	---	-----	-----------	---

R6	6	106	30-SEP-26	2
----	---	-----	-----------	---

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

R7	7	107	01-OCT-26	5
----	---	-----	-----------	---

R8	8	108	02-OCT-26	4
----	---	-----	-----------	---

R9	9	109	03-OCT-26	3
----	---	-----	-----------	---

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
-----------	-------------	------------	-------------	--------

R10	10	110	04-OCT-26	5
-----	----	-----	-----------	---

R10	10	110	04-OCT-26	5
-----	----	-----	-----------	---

10 rows selected.

Display Customer Names with Their Reviews: Displays customer names along with the artwork IDs, ratings, and review comments. This helps identify the feedback provided by individual customers.

```
SQL> SELECT c.Customer_Name, r.Artwork_ID, r.Rating, r.Review_Comments FROM Customer c JOIN Review r ON c.Customer_ID = r.Customer_ID;
```

CUSTOMER_NAME	ARTWORK_ID	RATING
REVIEW_COMMENTS		
Dhoni	101	5
Beautiful artwork with excellent colors		
Virat	102	4
Good artwork with attractive colors		
Ashwin	103	3
Good portrait with decent detailing		

CUSTOMER_NAME	ARTWORK_ID	RATING
REVIEW_COMMENTS		
SanjuSamson	104	5
Excellent charcoal artwork with fine details		
Shreyas Iyer	105	4
Creative digital artwork with vibrant colors		
Hardhik Pandya	106	2
Artwork is good but needs more detailing		

CUSTOMER_NAME	ARTWORK_ID	RATING
REVIEW_COMMENTS		
Ruturaj Gaikwad	107	5
Beautiful painting with a peaceful atmosphere		
Rohit Sharma	108	4
Impressive sculpture with detailed craftsmanship		
Raina	109	3
Interesting abstract design with good colors		

Find Products Having Maximum Reviews: Identifies the artworks that have received the highest number of customer reviews.

```
SQL> SELECT Artwork_ID, COUNT(*) AS Total_Reviews FROM Review GROUP BY Artwork_ID HAVING COUNT(*) = (SELECT MAX(COUNT(*)) FROM Review GROUP BY Artwork_ID);
```

ARTWORK_ID	TOTAL_REVIEWS
101	1
102	1
105	1
109	1
103	1
108	1
110	1
106	1
104	1
107	1

10 rows selected.

Display Recent Customer Feedback: Arranges reviews according to their review dates, displaying the most recent feedback first.

```
SQL> SELECT * FROM Review ORDER BY Review_Date DESC;
```

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
REVIEW_COMMENTS				
R10	10	110	04-OCT-26	5
Beautiful handmade craft with excellent finishing				
R9	9	109	03-OCT-26	3
Interesting abstract design with good colors				
R8	8	108	02-OCT-26	4
Impressive sculpture with detailed craftsmanship				

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
REVIEW_COMMENTS				
R7	7	107	01-OCT-26	5
Beautiful painting with a peaceful atmosphere				
R6	6	106	30-SEP-26	2
Artwork is good but needs more detailing				
R5	5	105	29-SEP-26	4
Creative digital artwork with vibrant colors				

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING
REVIEW_COMMENTS				
R4	4	104	28-SEP-26	5
Excellent charcoal artwork with fine details				
R3	3	103	27-SEP-26	3
Good portrait with decent detailing				
R2	2	102	26-SEP-26	4
Good artwork with attractive colors				

Retrieve Reviews with Ratings Above 4: Identifies reviews with ratings greater than 4, helping highlight highly positive customer feedback.

```
SQL> SELECT * FROM Review WHERE Rating > 4;
```

REVIEW_ID	CUSTOMER_ID	ARTWORK_ID	REVIEW_DATE	RATING	REVIEW_COMMENTS
R1	1	101	25-SEP-26	5	Beautiful artwork with excellent colors
R4	4	104	28-SEP-26	5	Excellent charcoal artwork with fine details
R7	7	107	01-OCT-26	5	Beautiful painting with a peaceful atmosphere
R10	10	110	04-OCT-26	5	Beautiful handmade craft with excellent finishing

7. Calculate Average Product Ratings

Aggregate functions are used to analyze the ratings received by different artworks.

Calculate Average Rating for Each Product: Determines the average customer rating for each artwork based on its available reviews.

```
SQL> SELECT Artwork_ID, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID;
```

ARTWORK_ID	AVERAGE_RATING
101	5
102	4
105	4
109	3
103	3
108	4
110	5
106	2
104	5
107	5

10 rows selected.

Count the Number of Reviews per Product: Calculates the total number of reviews received by each artwork.

```
SQL> SELECT Artwork_ID, COUNT(*) AS Total_Reviews FROM Review GROUP BY Artwork_ID;
```

ARTWORK_ID	TOTAL_REVIEWS
101	1
102	1
105	1
109	1
103	1
108	1
110	1
106	1
104	1
107	1

10 rows selected.

Find Highest-Rated Products: Identifies the artwork or artworks with the highest average rating.

```
SQL> SELECT Artwork_ID, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID HAVING AVG(Rating) = (SELECT MAX(AVG(Rating)) FROM Review GROUP BY Artwork_ID);
```

ARTWORK_ID	AVERAGE_RATING
101	5
110	5
104	5
107	5

Identify Products with Average Rating Above 4: Displays artworks whose average ratings are greater than 4, indicating high customer satisfaction.

```
SQL> SELECT Artwork_ID, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID HAVING AVG(Rating) > 4;
```

ARTWORK_ID	AVERAGE_RATING
101	5
110	5
104	5
107	5

These calculations help compare artwork performance and understand customer opinions.

8. Identify Highly Rated Products

The system generates reports to evaluate artwork ratings and customer feedback.

Report 1: Product Rating Analysis

This report displays the artwork name, number of reviews, and average rating. It helps compare artworks based on customer feedback and identify those with better ratings.

```
SQL> SELECT Artwork_ID, COUNT(*) AS Total_Reviews, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID ORDER BY Average_Rating DESC;
```

ARTWORK_ID	TOTAL_REVIEWS	AVERAGE_RATING
104	1	5
101	1	5
107	1	5
110	1	5
102	1	4
105	1	4
108	1	4
109	1	3
103	1	3
106	1	2

10 rows selected.

Report 2: Customer Feedback Analysis

This report focuses on three important areas:

- **Most Reviewed Products:** Identifies artworks receiving the greatest number of customer reviews.

```
SQL> SELECT Artwork_ID, COUNT(*) AS Total_Reviews FROM Review GROUP BY Artwork_ID ORDER BY Total_Reviews DESC;
```

ARTWORK_ID	TOTAL_REVIEWS
101	1
102	1
105	1
109	1
107	1
108	1
110	1
106	1
104	1
103	1

10 rows selected.

- **Highly Rated Products:** Identifies artworks that receive high average ratings.

```
SQL> SELECT Artwork_ID, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID HAVING AVG(Rating) >= 4 ORDER BY Average_Rating DESC;
```

ARTWORK_ID	AVERAGE_RATING
104	5
110	5
101	5
107	5
105	4
102	4
108	4

7 rows selected.

- **Products Requiring Improvement:** Highlights artworks with low average ratings that may require improvements in quality, presentation, or customer experience.

```
SQL> SELECT Artwork_ID, AVG(Rating) AS Average_Rating FROM Review GROUP BY Artwork_ID HAVING AVG(Rating) < 3;
```

ARTWORK_ID	AVERAGE_RATING
106	2

Report 3: Rating Distribution

This report analyzes how customer ratings are distributed across the available artworks.

- **Number of 5-Star Ratings:** Counts reviews that received the highest rating.

```
SQL> SELECT COUNT(*) AS Five_Star_Ratings FROM Review WHERE Rating = 5;
```

FIVE_STAR_RATINGS
4

- **Number of 4-Star Ratings:** Counts reviews that received a rating of 4.

```
SQL> SELECT COUNT(*) AS Four_Star_Ratings FROM Review WHERE Rating = 4;
```

FOUR_STAR_RATINGS
3

- **Number of Low-Rated Reviews:** Counts reviews with ratings of 1 or 2 stars.

```
SQL> SELECT COUNT(*) AS Low_Rated_Reviews FROM Review WHERE Rating <= 2;
```

LOW_RATED_REVIEWS
1

- **Complete Rating Distribution:** Displays the number of reviews received for each rating from 1 to 5 stars.

```
SQL> SELECT Rating, COUNT(*) AS Total_Reviews FROM Review GROUP BY Rating ORDER BY Rating DESC;
```

RATING	TOTAL_REVIEWS
5	4
4	3
3	2
2	1

The rating distribution helps the marketplace understand overall customer satisfaction and identify areas that may need attention.

9. Commit and Verify the Records

After completing the database operations, the changes are committed to save pending transactions. The Review table is then displayed to verify the stored review records, including review IDs, customer IDs, artwork IDs, dates, ratings, and comments.

This verification helps ensure that the review information has been stored correctly.

10. Result

The Product Review and Rating Management System was developed successfully for the ArtHub online art marketplace. The Review table stores customer feedback and ratings and maintains relationships with the existing Customer and ArtProduct tables.

The system supports review retrieval, rating calculations, review counts, identification of highly rated artworks, and rating distribution analysis. The generated reports provide useful information about customer satisfaction and artwork performance.

11. Conclusion

The Week 6 DBMS exercise demonstrates the use of SQL for managing customer reviews and product ratings in an online marketplace. The system maintains review records, applies database constraints, and generates reports to analyze customer feedback.

The information obtained from these reports helps ArtHub understand customer preferences, recognize highly rated artworks, and identify artworks that may require improvement. Thus, the Product Review and Rating Management System supports better customer satisfaction and effective artwork management.